

Amazon deforestation implications in local/regional climate change

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Amazonia and Global Climate Change: A two-way process



Deforestation and forest degradation due to climate change

Fig. 1. Amazonia and global climate change: a two-way process, where emissions from deforestation impact greenhouse gas emissions and climate change. However, the increase in temperature and reduced precipitation is causing forest degradation, enhancing carbon emissions.

Amazonia and the tropical forests are full of scientific surprises, with important environmental and climatic impacts that range from local to regional and global. Before the Industrial Revolution, there was quite an equilibrium between forest photosynthesis and ecosystem respiration, keeping the carbon stock of tropical forests in equilibrium (1). However, the increase in atmospheric carbon dioxide concentrations caused forests to absorb more carbon than they released, with the Global Carbon Project estimating that forests are absorbing 26% of the CO₂ emissions (2). The Amazon rainforest is highly biodiverse and the largest contiguous tropical rainforest in the world. The links between forest and climate are very strong regarding precipitation, temperature, radiation, aerosols, clouds, and other variables (3). Carbon cycling is very much coupled to climate and water cycles in

tropical rainforests. Undisturbed forests act as a carbon sink by taking up about 11.4 GtCO₂/y, or 29% of anthropogenic emissions annually, while v accounts for an emission of 4.5 GtCO₂/y, or 11% of emissions (2). However, deforestation

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and climate change are altering tropical forest's important ecosystem services. Tropical deforestation is an important driver of global climate change through emissions of greenhouse gases. But the reverse is also important, where increased temperatures, reduced precipitation, and increased climate extremes, such as droughts, are making some regions in the Amazon Forest start to have a positive net carbon flux to the atmosphere (4, 5).

Fig. 1 illustrates this new chain of impacts, with deforestation greenhouse gas emissions feeding the increase in global climate change, while the increased temperature and reduced precipitation associated with climate change alter the forest metabolism, triggering forest degradation that further enhances emissions and carbon loss by the ecosystem. Amazonia alone has about 120 Pg of carbon stored in the ecosystem (6), equivalent to about ten years of fossil fuel emissions. We must avoid transforming this land carbon stock into atmospheric CO₂, aggravating the already severe climate change.

The study quantifies the critical contribution of tropical deforestation to regional climate warming and the potential for reduced deforestation to deliver regional climate adaptation and resilience with important implications for the sustainable management of the Amazon.

Tropical deforestation impacts the climate through complex land-atmosphere interactions causing local and regional warming (7, 8). The local effects of deforestation, with increases in local temperature, reducing evapotranspiration, and altering surface albedo, are relatively well documented (3, 5, 6). But at the regional level, what could be the effect of deforestation in Amazonia? This was the main issue addressed by Butt et al. (9) in an important study. The biophysical effects of forests can strongly impact the properties of the land surface through changes in land-atmosphere fluxes of heat, moisture, and momentum (3, 5). They used remotely sensed observations of forest loss and dry season land-surface temperature measurements to demonstrate that deforestation of the Amazon caused strong warming at distances up to 100 km from the forest loss. They apply a machine-learning approach using extensive remote-sensing information. It calculated the change in remote-sensing observations of the land surface at 3.7 million locations, each one km² in extent across the Amazon between 2001 and 2020. They explored how warming from forest loss depended on the size of both local and nonlocal forest loss. The machine learning approach was used to further isolate forest loss's local and nonlocal effects. The constructed model was then used to make the first prediction of how temperature response depends on local and nonlocal forest loss across the Amazon. They showed that nonlocal warming due to forest loss at 2 to 100 km length scales increases the warming due to deforestation by more than a factor 4, from 0.16 K to 0.71 K for each 10-percentage point of forest loss. In addition, they have done future land-use scenarios to estimate that deforestation could cause additional warming of 0.5 to 0.6 K across the Brazilian Amazon biome from 2020 to 2050. Global warming, coupled with this effect, could potentially facilitate the Amazon Forest to reach a tipping point (10), where the

forest could lose significant amounts of carbon to the atmosphere, significantly aggravating climate change (11, 12). Several regions in the Amazon have increased in temperature by about 2 Celsius and have undergone significant decreases in precipitation (13). With the projected warming along this century, we could potentially reach the tipping point of Lovejoy and Nobre (10), promoting a significant loss of carbon by the ecosystem, with impacts on the global climate (14).

The study quantifies the critical contribution of tropical deforestation to regional climate warming and the potential for reduced deforestation to deliver regional climate adaptation and resilience with important implications for the sustainable management of the Amazon. Indeed, the Amazonian Forest degradation observed in several recent studies and the consequent net carbon emissions to the atmosphere are also linked to this effect of regional deforestation (4, 11, 15).

It is also important to notice that this observed regional increase in temperature contributes to forest degradation. In the past decades, a declining trend in the carbon sink capacity in the Amazon rainforest has been observed due to increased carbon losses and tree mortality (16). Rising temperatures and more frequent severe droughts are potentially major drivers. Severe droughts such as the ones in 2005 and 2010 have turned the Amazon carbon sink into a temporary carbon source (17, 18). A recent study (16) shows that Amazonian, African, and Southeast

Asia forests could be close to a tipping point, where they could start losing carbon to the atmosphere. The critical temperature beyond which photosynthetic machinery in tropical trees begins to fail averages approximately 46.7 °C (11). However, it remains unclear whether Leaf photosynthesis thermal limits will be reached soon or will under climate change (15). Humid tropical rainforests are an ecosystem being impacted hardly by climate change. As temperatures in tropical forests are near or above the temperature optimum for photosynthesis, further increased temperatures may close stomata, reducing transpirational cooling and exposing leaves to damaging temperatures. The paper also shows that previous estimates of warming associated with deforestation are underestimated since they do not fully account for the contribution of regional forest loss. They conclude that a tipping point in metabolic function in tropical forests could occur with 3.9 °C of additional warming, which is expected for tropical forests under IPCC high emissions scenarios (19). These climate scenarios must be avoided to keep healthy tropical forests. In addition, deforestation and fragmentation can amplify local temperature changes (8). Climate change and local deforestation may already place the hottest tropical forest regions close to, or even beyond, a critical thermal threshold (12). Notably, Butt et al., paper focuses on the analysis of the dry season. Possibly, these effects should exist across all seasons. However, future work is needed to assess the nonlocal climate impacts outside the dry season.

Modeling this effect is undoubtedly difficult. Tropical forests are a challenge for earth system models (ESM). Most ESMs project a continuous increase in the tropical carbon sink in the future, mainly due to the plant-physiological effects of increased CO₂ concentrations (20). The diversity of tropical tree species in CMIP6 models is often only represented in

single plant functional type parameterizations, which is unlikely to capture the correct response of plants under drought stress, CO₂ fertilization, on carbon gains, albeit limited by nutrient availability in some models outweigh negative climate impacts in future simulations (21). Possibly, the unrealistic representation of forest drought response in process-based vegetation models may lead to this mismatch. So far, most dynamic global vegetation models cannot show spatial patterns associated with drought stress and do not show a significant declining trend in net carbon sink rates. We are a pretty long way to a realistic representation in models of the complexity of tropical forests' responses to climate change, with the large biodiversity and diverse climatic conditions they contain. Additionally, several nonlinear responses between the multiple drivers (such as leaf temperature, maximum climatological water deficit, surface energy balance, water and carbon fluxes, nutrient availability, vapor pressure deficit, and other parameters) make this complex issue a challenging. The role of drought stress and increased temperature in the long-term decline of the Amazon carbon sink is still unclear. Integrated efforts involving climate and tropical forests' biological functioning are necessary to understand this critical issue better. The sensitivity to temperature increase and resilience to drought is species-dependent, so integrating biodiversity would also be important, as emphasized by the joint IPBES and IPCC Report on Climate and Biodiversity (22).

There is good news. Brazil announced in September 2023 that deforestation in the Amazon had been reduced by 57% compared with a similar period in 2022. Unfortunately, public and indigenous land-grabbing and illegal mining are bringing the illicit activities to a center stage in Amazonia. It will not be easy, but feasible, to get zero deforestation by 2030, as committed by Brazil in his NDC (Nationally Determined Contribution) of the Paris Agreement. But it is essential to do so for South America's and the planet's climate stability (3). A joint effort between South America, Africa, and Southeast Asia

tropical forest countries is important to jointly reduce tropical forest greenhouse gas emissions. We must also enhance carbon sink in these degraded areas through large-scale ecological restoration projects.

These discussions highlight the role of deforestation in regional and global climate change and emphasize the importance of reducing deforestation for climate adaptation and resilience in the Amazon. Several recent works mention that tropical forests could be close to a tipping point, where the hydrological cycle that sustains these tropical forests could be severely perturbed and bring the forests to a different ecosystem, able to store much less carbon. Identifying options for sustainable development for the region is essential to support climate adaptation and resilience for Amazonia, and this needs to be done considering the needs of local "river people" and indigenous communities. Similar Amazonian climatic mechanisms are possibly occurring in the tropical forests of Africa and Southeast Asia, and more work is needed to explore the nonlocal impacts of deforestation in other forested regions. Climate and atmospheric circulation are different for all tropical rainforests, so results from this study should not be applied to other tropical forests without further studies. It is essential to have a global approach to preserve tropical forests and their significant carbon stocks. Reducing deforestation is undoubtedly the easiest, cheaper, and quicker way to reduce greenhouse gas emissions, with substantial cobenefits of preserving ecosystem services. The results from Butt et al. demonstrate the contribution of tropical deforestation to regional climate warming and the potential for reduced deforestation to deliver regional climate adaptation and resilience with important implications for sustainable management of the Amazon and global tropical forests.

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